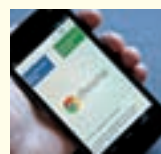


TWEAKING MUSIC FOR THE IPOD AGE

Audio engineers are increasingly creating specially mastered versions of songs designed to counteract the audio degradation caused by compression. Mastering engineers are doing their best to pass through Apple's iTunes algorithms with minimal sonic corruption
<http://goo.gl/dP3yB>

BEGONE CRAPWARE!

Many of us who buy new desktops or laptops are surprised by the number of programs that litter our screen. We call that bloatware. Find out if you need or want it, and ways to remove if you don't. PC Mag presents a step-by-step tutorial on identifying and removing these obnoxious software
<http://goo.gl/hSVGi>

CHROME FOR ANDROID

Google is finally making the move we've been expecting

for some time now and bringing Chrome to Android. Engadget presents a hands-on video of Chrome for Android running in ICS on a Samsung Galaxy Nexus
<http://goo.gl/4HcGg>

GOOGLE OR Groupon

Facebook is finally going public. It's either going to be the next biggest thing on Wall Street since money or Facebook's IPO will meet a fate similar to that of this year's other high-profile tech IPOs like Zynga and Groupon. In sync with our cover story here's Wired's take on the same
<http://goo.gl/Hbkzc>

The state of DDR4

Truths about the next big thing for RAMs, and how fast we'll see it sitting in our computers...

- By Matthew Murray



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Do almost anything often enough, and the individual steps become routine. It certainly feels that way when I'm building computers: There are days I feel like I could do it in my sleep (and, truth be told, there are days I probably have). But when you're selecting parts, most of the components require at least a little analysis. The CPU and the motherboard are major, and have to be considered together. Of course storage is a big deal — you always want enough and you always want it to be fast, and optimizing those qualities with affordability can be a headache (especially at the moment [1]). For a gaming system, the right video card is essential. And in addition to being your system's face to the world, the case will affect both the way you build the computer and how much you can expand it later. And... that's it. Right? Am I forgetting anything?

Oh yeah, RAM. Sorry about that. But I admit it: When I'm shopping for memory, I pretty much turn off my brain. It's not intentional, and it's not that I think RAM isn't important. It's just that it's the easiest kind of hardware to shrug off. Most of us, even when we're in our most rabid enthu-

siast mode, roll our eyes at the arcane industry of RAM timings. Provided the DIMMs' voltages and speeds are compatible with the motherboard, everything else usually seems academic. Oh, if I'm being super picky about components, I might pay a bit more attention, but in reality that's rarely necessary. These days, you can take it for granted that basic DDR3 is going to be what you want.

It occurred to me last week, however, that we're rapidly approaching another time when that might not be the case. As processors and other hardware get more powerful, it becomes more difficult — and important — for RAM to keep up. Over the last 20 years or so we've seen single data rate (SDR) memory give way to double data rate (DDR) memory, which transfers data on both the rising and falling edges of the clock signal. And over the last decade,

we've slowly progressed from DDR to DDR2 (with an internal clock running at half DDR's rate) to DDR3 (operating at half DDR2's rate), constantly upping performance while reducing power usage along the way. But what about the next logical step in this particular cycle: DDR4?

It's coming, though when — and whether — you'll see it for use in your home computer is still somewhat up in the air.

Samsung announced a year ago that it had developed the first 2GB DDR4 modules using a 30nm process technology, achieving with them transfer rates of 2.133Gbps at 1.2 volts. Three months later, Hynix came up with its own DDR4-2400 modules operating at the same voltage.

The website DDR4.org, which is (not surprisingly) devoted to information about the upcoming memory standard,

is a bit light on substantive information but fleshes out a few of the details. Among them: Data transfer rates would start at 2,133MT/s (about where DDR3 is leaving off) and could eventually double to 4,266MT/s, and initial energy consumption is expected to be about 1.2 volts (though later chips might be able to use as little as 1.05 volts). The site goes on to say that "other sources

